

Literature Survey

Combinatorial Optimization on Restricted Graph Classes

Scope

Problems: Maximum Independent Set / Minimum Vertex Cover, Dominating Set and variants, Matching and related variants.

Graph classes: P_t -free graphs, interval graphs, chordal graphs, chordal bipartite graphs, bounded-treewidth graphs, bounded-degree graphs, and minor-free graphs.

Primary focus: Complexity boundaries — cases where restricting the graph structure changes a problem from NP-hard to polynomial-time solvable, or where the complexity remains unresolved.

1. Maximum Independent Set / Vertex Cover on P_t -free Graphs

Maximum Independent Set (MIS) and Minimum Vertex Cover (MVC) are complementary:

$$\alpha(G) + \tau(G) = |V(G)|.$$

Hence, complexity results for one immediately transfer to the other.

Known Complexity

Graph Class	Status
P_4 -free	Polynomial-time
P_5 -free	Polynomial-time
P_6 -free	Polynomial-time
P_7 -free with bounded clique number	Polynomial-time (2025)

Graph Class	Status
General P_7 -free	Open
General P_t -free, fixed $t \geq 7$	Open complexity frontier

Polynomial-time algorithms are known through P_6 -free graphs. A recent result extends polynomial-time solvability to P_7 -free graphs when the clique number is bounded.

The complexity of MIS on unrestricted P_7 -free graphs remains unresolved. More generally, no fixed t is currently known for which MIS becomes NP-hard on P_t -free graphs.

Structural Boundary

A significant part of the recent work involves:

- minimal separators;
- potential maximal cliques;
- separator-covering techniques.

The current boundary can be viewed as:

$$P_6\text{-free} \longrightarrow P_7\text{-free} + \text{additional restrictions} \longrightarrow P_7\text{-free}.$$

Possible Research Direction

Rather than approaching unrestricted P_7 -free graphs directly, consider natural subclasses

$$(P_7, H)\text{-free}$$

for suitable small graphs H , or P_7 -free graphs satisfying another structural restriction.

Related Problem

Connected Vertex Cover on P_6 -free graphs also has unresolved complexity.

2. Domination on Interval and Chordal Graphs

For a graph $G = (V, E)$, a dominating set $D \subseteq V$ satisfies

$$\forall v \in V \setminus D, \quad N(v) \cap D \neq \emptyset.$$

A $[1, j]$ -dominating set additionally requires

$$1 \leq |N(v) \cap D| \leq j \quad \forall v \in V \setminus D.$$

The upper bound on the number of dominating neighbours substantially changes the algorithmic behaviour of the problem.

Complexity Boundary

Graph Class	Problem	Status
Interval	Dominating Set	Polynomial-time
Chordal	Dominating Set	NP-hard
Interval	$[1, j]$ -Domination, $j \geq 3$	Polynomial-time
General interval	$[1, 2]$ -Domination	Open
Non-proper interval	$[1, 2]$ -Domination	Polynomial-time (2025)
Circle	$[1, 2]$ -Domination	NP-complete

The $j = 2$ case therefore gives a particularly narrow unresolved boundary.

Recent Development

A 2025 result gives a polynomial-time dynamic-programming algorithm for $[1, 2]$ -domination on **non-proper interval graphs**.

The general interval-graph case remains unresolved.

Natural classes between the known and unknown cases include:

- unit interval graphs;
- proper interval graphs;
- other structurally restricted interval graphs;

- general interval graphs.

Possible Research Direction

Determine whether the structural ideas or dynamic-programming formulation used in the restricted case can be extended to a larger interval-graph class.

Central Question

Which structural property prevents the existing algorithm from extending to general interval graphs?

Identifying this obstruction may lead either to an extended polynomial-time algorithm or to a sharper complexity boundary.

3. Dominating-Set Enumeration on Chordal Bipartite Graphs

A related direction concerns **enumeration complexity**.

Instead of finding one dominating set, the objective is to enumerate all minimal solutions while controlling the delay between consecutive outputs.

Current Boundary

Graph Class	Problem	Best Known Result
Bipartite	Minimal Dominating Sets	Polynomial-delay status open
Chordal bipartite	Minimal Dominating Sets	Polynomial delay (2025)
Chordal bipartite	Minimal Total Dominating Sets	Polynomial delay (2025)
Chordal bipartite	Minimal Connected Dominating Sets	Incremental-polynomial
Chordal bipartite	Minimal Connected Dominating Sets	Polynomial delay open

Recent work therefore resolves the ordinary and total variants on chordal bipartite graphs while leaving the connected variant at a weaker enumeration-complexity bound.

Possible Research Direction

Polynomial-delay enumeration of minimal connected dominating sets on chordal bipartite graphs, potentially beginning with further restricted subclasses.

4. Other Complexity Boundaries

4.1 Independent Set on Bull-Free Graphs

- **Claw-free**: polynomial-time.
- **Fork-free**: polynomial-time.
- **Bull-free**: polynomial Turing kernel known; polynomial-time solvability remains unresolved.
- **(Bull, odd-hole)-free**: polynomial-time.

Natural subclasses such as (bull, H) -free graphs provide possible intermediate cases.

4.2 Efficient Domination on P_t -free Graphs

- General P_7 -free graphs — NP-complete.
- P_7 -free bipartite graphs — polynomial-time.
- P_8 -free bipartite graphs — polynomial-time.
- General P_8 -free and P_9 -free bipartite cases contain unresolved boundaries.

4.3 Bounded Treewidth / Minor-Free Graphs

Many classical optimization problems already have strong parameterized algorithms and tight complexity bounds on bounded-treewidth graphs.

Minor-free graph algorithms provide further open directions, particularly for domination variants, but require substantially more machinery from parameterized complexity and structural graph theory.

5. Candidate Research Directions

I. $[1, 2]$ -Domination on Interval Graphs

Current Boundary

restricted interval classes $\longrightarrow$ polynomial

while

general interval $\longrightarrow ?$

Tools required:

- interval representations;
- vertex orderings;
- dynamic programming.

Possible progression:

restricted interval class $\longrightarrow$ proper/unit interval $\longrightarrow$ general interval.

II. Maximum Independent Set on Restricted P_7 -free Graphs

Current Boundary

P_6 -free $\longrightarrow P$

P_7 -free + bounded $\omega \longrightarrow P$

P_7 -free $\longrightarrow ?$

Tools required:

- structural decomposition;
- minimal separators;
- potential maximal cliques.

Possible progression: study (P_7, H) -free graphs or other restricted subclasses before approaching general P_7 -free graphs.

III. Connected Dominating-Set Enumeration on Chordal Bipartite Graphs

Current Boundary

Minimal DS $\rightarrow$ polynomial delay

Minimal Total DS $\rightarrow$ polynomial delay

Minimal Connected DS $\rightarrow$ incremental polynomial only.

Tools required:

- enumeration algorithms;
- structural properties of chordal bipartite graphs;
- neighbourhood decomposition.

6. Current Preferred Direction

Research Anchor

$[1, 2]$ -Domination on Interval Graphs

Why this direction?

- **Clearly defined unresolved complexity boundary**
- Recent result provides a concrete starting point
- Interval graphs have accessible structural representations
- Dynamic programming provides an identifiable algorithmic framework
- Intermediate subclasses allow the problem to be narrowed further
- Lower prerequisite barrier than the P_7 -free MIS frontier

Immediate Questions

1. What structural property is used by the existing polynomial-time algorithm?
2. Exactly where does the argument fail for a more general interval representation?
3. Does the same obstruction occur for proper or unit interval graphs?
4. Can the state space of the dynamic program be modified to handle the missing configurations? If not, does the obstruction suggest a hardness construction?